

17/03/18.

UNIT-3.

POLLUTION

①

I. Water Pollution

Sources : • Natural. (i) decomposition of organic matter.
(ii) soil erosion.

• Human Activity

(i) Industrial waste (ii) Agriculture waste

→ O_2 demanding Waste.

Decrease DO levels. This includes mainly the organic waste from industries and domestic lands.

→ Pathogens: Disease causing organisms coming into the water through sewage waste from tanning, leather food processing industries.

Two types of diseases:

• Water born disease • Water contact disease.

→ Synthetic Organic Compounds.

• Pesticide • Detergents.
 ← Surfactant (30%)
 Builders (70%)

Builders cause eutrophication due to high nutrient content and surfactant make the water turbid and full of lather.

→ Plant Nutrients

Sources are industrial waste and surface runoffs. They reduce the dissolved O_2 and cause eutrophication.

→ Pollutant: Inorganic chemicals and minerals.

(i) Mineral acids: comes in water due to acid mine drainage.

(ii) Soluble salts: It increases TDS of water and after certain permissible limit kills the aquatic life.

(iii) Heavy Metals: sources: • Thermal power plants
• Industrial effluents.

effect: • Cause nervous system damage.
• Cancer.
• ~~Mutations~~ Mutation
• Kidney damage.

→ Sediments: Includes:- • sand • soil • silt • Rock particles
which increases turbidity and blocks → sunlight.
Reduces photosynthesis and dissolved O_2 .

Causes Effects: • Asphyxiation (Breathlessness) for aquatic animals.

→ Radio-active Substances

sources: • Mining of radio-active substance.
• Nuclear power plants.
• Industrial use.
• Medicinal and research purpose.
• Cosmic rays. • Production & testing of nuclear weapons.

effects: Through food it enters into human bodies and get accumulated in blood and organs. It causes Leucemia (blood cancer), mutation, bone cancer and eye's defects.

→ Thermal Discharges

Sources:- • All industries which use water as coolant ~~and~~ contribute to thermal discharges.

Example:- • Steel industries • Atomic energy plants.

Effects:- • It rises water temp. which depletes DO levels.

• Kills the most essential enzymes for life.

→ Oil

Sources:- • Oil refineries • Industrial effluents.
• automobile waste oil • Ballast water.

Effects: • Stops aeration of water body.

→ Volatile Organic Compound (VOC)

②

These are used as solvents in most chemical industries but they are very toxic.

example: • Vinyl chloride → cancer.
• CCl_4 → Poisonous

→ Flocculation

① Pretreatment → involves screening and grit removal

W.W → screens → comminutors → grit chamber

② Primary Sedimentation

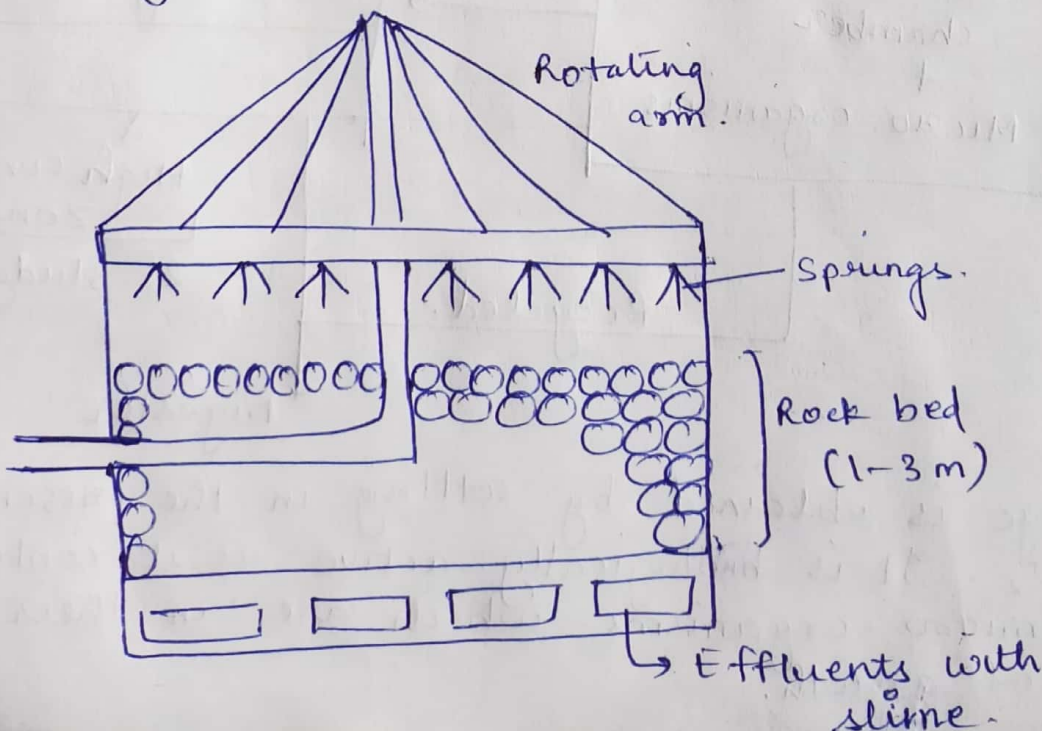
Primary sedimentation

I. Primary Treatment

II Secondary Treatment

It is used to remove all the O_2 demanding waste which is in the soluble or colloidal forms by using bio-logical processes.

(i) Flicking Filters / Biological Towers.



Individual rock is covered with slime layer, which consist aerobic micro-organisms, when the influents are sprinkled over the rock bed, it trickles over the slime layer and the dissolved organic matter moves inside this slime layer where it gets decomposed aerobically and produce CO_2 and H_2O , which comes out with the effluents initially but as the slime layer thickness it gets detached from the rocks and comes out in the effluent in the form of humous which can be collected as sludge sent to the digesters and finally used as manure.

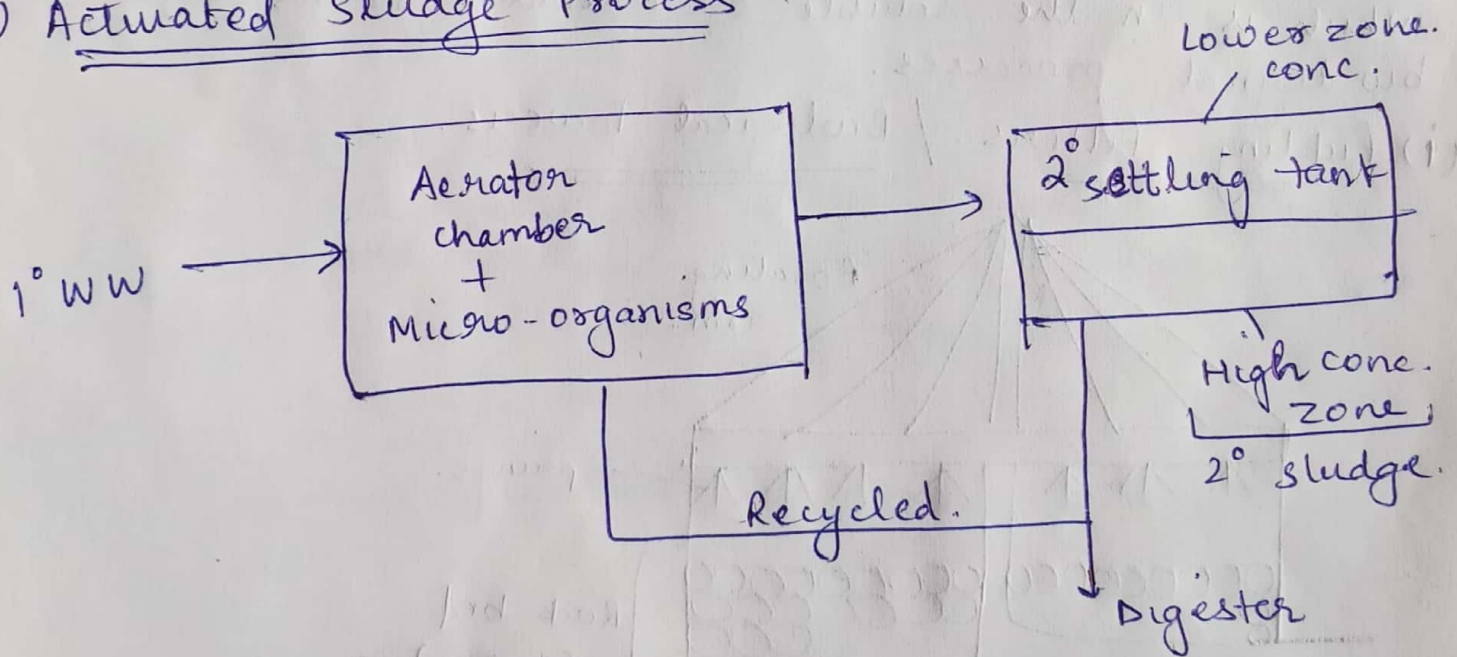
Advantages :-

- Reduces the BOD. ~~of water~~ about 60-80%.
- It has low operating cost.
- constant monitoring is not required.
- effluents are of better quality.

Drawbacks :-

- Initial cost is high.
- Whole process is sensitive to temp, p^{H} , etc.

(ii) Activated Sludge Process



Activated sludge is obtained by settling in the presence of excess of O_2 . It is biologically active as it contains high conc. of micro-organisms which are in their active state of growth.

Advantages : • Cheap process.

• Reduces BOD

• Small area required.

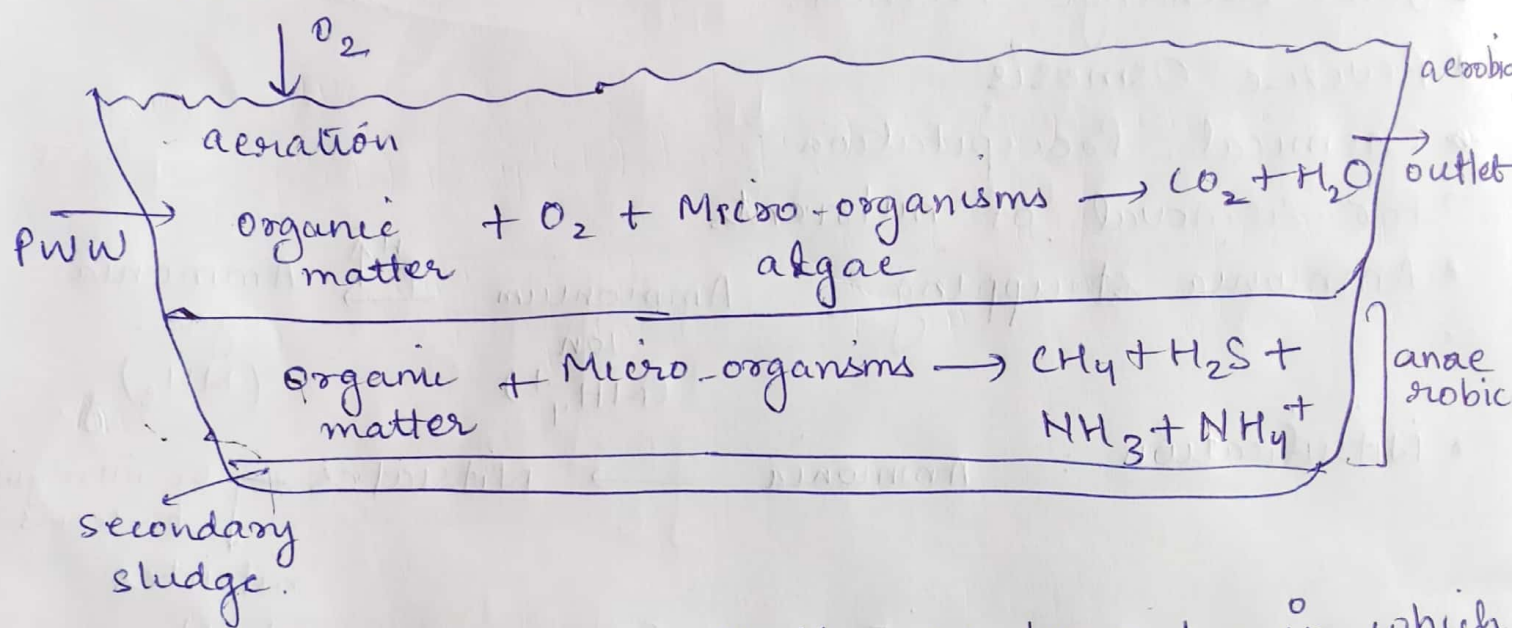
Drawbacks : • 1-5 ppm O_2 has to be maintained

• Constant monitoring required.

• sensitive to high temp. and pH values.

• Not able to remove non-biodegradable organic matter.

(iii) Oxidation Ponds



These are the large but shallow ponds only in which the waste water entered through one side and the organic matter collected through other side. The organic matter decomposition or upper zone is aerobic whereas near the bottom anaerobic decomposition takes place which produces bad odour gases. Therefore they are also called Facultative ponds.

Advantages & Disadvantages.

III Tertiary treatment

→ For removal of finer solids:

- Micro-straining

→ Removal of dissolved solids.

- Using activated Carbon filters for adsorption of soluble organic matter.

- Solvent extractions. → for phenolix

- Ion-Exchange Method.

- Reverse Osmosis

- Chemical Precipitation

→ For removal of nutrients

- Ammonia stripping: Ammonium ion (NH_4^+) $\xrightarrow{\text{pH} \uparrow}$ Ammonia gas $(\text{NH}_3)_g$.

- Nitrification: Ammonia $\longrightarrow$ Nitrates $\xrightarrow{\text{de-nitrifying bacteria}}$ nitrites + $\text{N}_2(g)$

- ~~Chemical~~ chemical precipitation → for removal of phosphorus.

→ Removal of micro-organisms

- Chlorination

- Retaining water in the lagoons for long duration.

SOIL POLLUTION

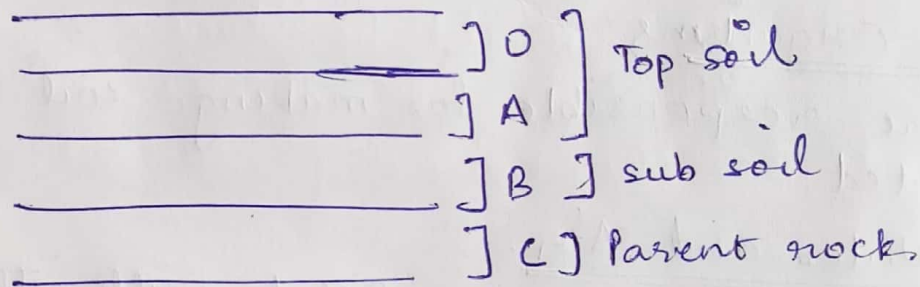
(4)

Soil is the top-most layer of earth's crust which are formed by weathering of rocks which contains inorganic minerals, organic matter, air, living organisms and water which together supports vegetation.

Factor for soil formation:

- Mechanical weathering of rocks due to temp changes, water, wind, ~~chemical~~
- Chemical weathering
- Biological weathering by Lichens.

Soil Profile: A cross sectional view of soil which shows different horizontal layers is called soil profile.



- ① O-horizon: It has freshly fallen leaves, dead organisms along with micro-organisms.
- ② A-horizon: It consists partially decomposed organic matter for humous along with high microbial population and worms.
- ③ B-Horizon: It consists of high inorganic matter, very less organic matter and negligible micro-organisms.
- ④ C-Horizon: It consists of weather parent material which helps in maintaining pH of soils.

Composition of soil

Different soil particles :-

- Clay < 0.02 mm
- Silt ≈ 0.002 mm
- Sand $\approx 0.02 - 0.2$ mm
- Gravel $0.2 - 2$ mm
- Stones 2 mm.

Inorganic matter.

- It includes • silicates of Na, K, Ca, Al
- Oxides of Fe and Mn.
 - Carbonates of Ca and Mg.
- These are also called plants nutrients.

Organic matter

It consists of • polysaccharides,

- nucleotides
- organo-sulphur compounds

• Rich in humous.

• Helps in maintaining soil porosity, water holding and retention capacity.

~~• Lignin.~~

Living Organisms

These are responsible for making soil fertile and well airted.

Soil Water and Air.

The pores of the soil are filled with either air or water which is controlling, regulating pH of soils as well as responsible for the oxidation of organic matter.

Soil Pollution

It is the contaminating of soil by chemical substance which changes its physical, chemical and biological properties and making it unfit for usage.

→ Sources

- Industrial waste → metals
- Urban waste. → plastics, glasses, metals, fibres
- Agricultural chemicals → fertilizers & pesticides
- Biological organisms → faulty sanitation, municipal waste
- soil erosion.
- coal based powerplants → fly & bottom ashes → heavy metals.

- Radio-active waste.

5

⇒ Effects of soil pollution

⇒ Control of soil pollution

- Waste should be dumped only after proper treatment.
- Toxic-persistent chemicals like pesticides should be banned instead bio-degradable substances should be used.
- proper awareness campaign should be run.
- crop rotation to be practiced.
- Bio-remediations and phyto-extractions should be done regularly for polluted soil.
- Afforestation programs should be run.
- Use of integrated pest management should be done.
- Organic farming, should be done.
- All sorts of organic waste should be segregated and utilized for the bio-gas production.

MARINE POLLUTION

Sources:-

- Rivers.
- Catchment areas. (coast. line with human settlement)
- oil drilling operations.

Pollutants :-

- oil.
- sewage (includes small solid particles)
- solid waste. (includes plastics, nylon ropes, nets) (cause respiratory choking).
- Chemicals
- Radio-active waste.
- Dissipating waste. → heat → depletes ~~the~~ DO levels.
→ acids & alkalis
- Eutrophication → The phyto-plankton population increases which have DMS in them which causes acidification.

CONTROL MEASURES

I. Source Level.

- Such technologies which generate no waste.
- avoid using toxic materials.

II. ~~Control~~ Disposal Level.

- Waste should always be treated first and then disposed off according to the set principles.

III. Legal Level.

- A specific licence or content should be required to start any production or developmental project.
- A detailed environment impact assessment (EIA) of the new discharges should be given in detail to stop or continue its production.

IV. Coastal Zone Management

- It controls the population activities
- Treatment and disposal strategies on coastal areas by working on EIA.

NOISE POLLUTION

Pollutants

- I. Natural :-
- storms.
 - Thunder.
 - Volcanic eruption
- II. Man made :-
- Domestic equipments
 - Industrial machinery
 - mining
 - Defense equipments
 - Entertainment devices.
 - Transport system.

Effects

I. Auditory effect

- Exposure to high intensity sound for short duration

causing temporary deafness.

- Permanent hearing loss.
- Decreased ability of ear. to respond to loud sounds.

II Non-Auditory

- Physiological disorders. → Anxiety, Insomnia.
- Increased irritability.
- Effect on wildlife

III Aquatic life

- The creatures use sounds for communication as well as navigation.
- High noises disrupt these processes.
- Migratory bird population is declining fast

Control

I Road Transport noise.

- Smoothening of roads.
- Using silencers.
- Avoid listening to music while driving.
- Avoid usage of horns.

II Air traffic Controls

- Engine designing to be improved.
- Altering path of flights away from population.

III Afforestation Programs.

- As the belts of trees are good absorber of sound

IV Legal Control

- Noise regulation rules are ammended under Environment protection Act 1986. which limits the noise levels sector wise.

ControlI. Legal Control

- Clean Water Act 1972
- Instead of discharging hot into water bodies immediately it should be passed through cooling towers and ponds, where they get cool down due to evaporation.
- Special artificial are constructed which can store heated water, where the water is cooled, recycled and reused.
- Heated water can be used as a source of heat for industrial and domestic purpose which is called ~~to~~ ~~cool~~ Co-generation.

THERMAL POLLUTION

Control

I. Legal Control

- Clean Water Act 1972
- Instead of discharging hot into water bodies immediately it should be passed through cooling towers and ponds, where they get cool down due to evaporation.
- special artificial **are** constructed which can store heated water., where the water is cooled, recycled and reused.
- Heated water can be used as a source of heat for industrial and domestic purpose which is called ~~to~~ ~~cool~~ Co-generation.

SOLID WASTE POLLUTION

- Sources :-
- From urban communities
 - Accumulation of agricultural, industrial and mining waste

Classification:-

- Garbage: It includes degradable waste with high moisture content approx 70%. with a heating value of 6×10^6 J/kg.

eg:- Left-over meat, vegetables & fruits.

- Rubbish:- It consist of non-degradable waste with 25% moisture and heating value of 15×10^6 J/kg. It is divided into combustible and non-combustible types.

- Pathological:- It includes dead animals and plants, disposable syringes, bandages, moisture content = 85%. with heating value = 2.5×10^6 J/kg. It is highly infectious waste. eg: Hospital waste.

- Industrial
- Agricultural

- Ashes
 - Light weight → Fly ash.
 - Heavy weight → Bottom ash.

Effects :- • It increases the disease causing organism and vectors.

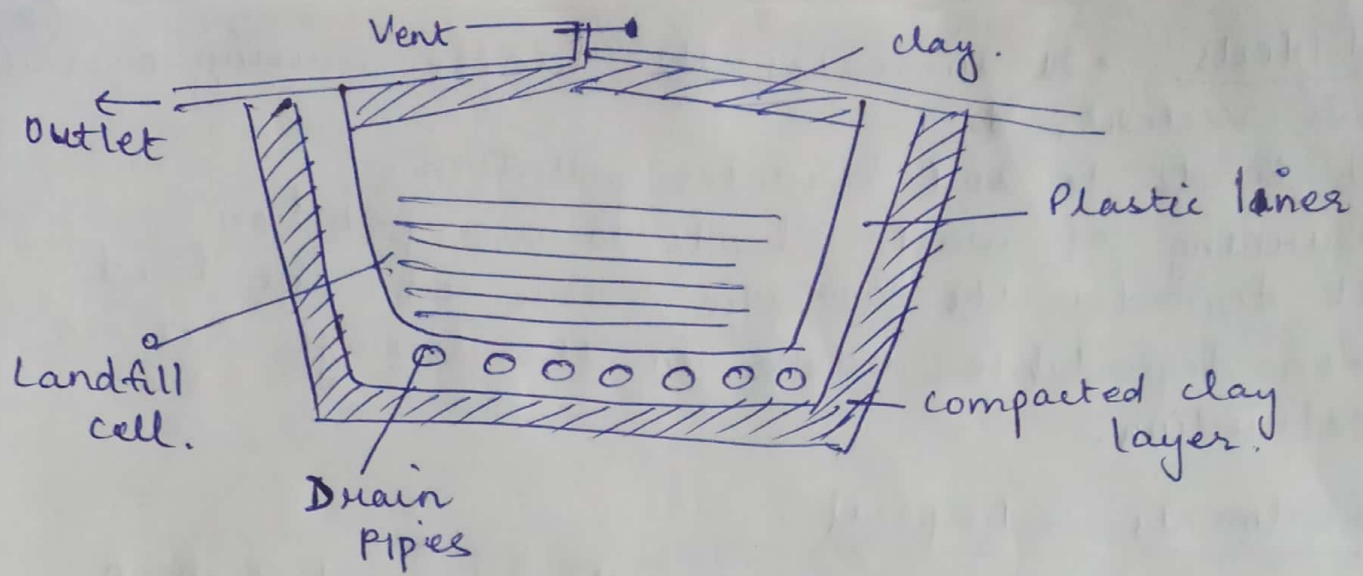
- It leads to soil & water pollution
- Burning of waste leads to air pollution.
- It degrades the aesthetic value of the land.
- Non-degradable waste in the sewage act as obstruction.

Treatments & Disposal :

- Dumping :- It requires no planning. It is cheap. where the waste is left untreated and uncovered and not being segregated. So they become breeding house for flies and mosquitoes and cause land and water pollution.
- Sanitary Land Fills :- The pit is dull, the garbage is dumped then the waste is covered with a soil layer. The layers are compressed and the new layer is added constantly which fills the entire cell. But when the water sweeps through it, it contaminates the surrounding soil and water. Therefore, they are supplied with 2 layers of filter the compacted clay layer followed by a plastic filter layer which is supplied by the drain-pipes to take out the lichens. Initially the decomposition is aerobic which utilizes all the oxygen and then the anaerobic decomposition starts which produces predominantly methane with the help of methane producing bacteria. Their activity slows down when the decomposition is about to end. and the aerobic conditions revert back.

Advantages :- • There is no infectious disease spread.

- Air, water & soil pollutions are prevented.
- As the structure is ~~not~~ supplied with proper vents connected with pipes, no fire hazard is possible.



Liquid Injection Incinerator:

It is the burning of waste in large furnaces which are supplied with excess O_2 and fuel to increase rate of combustion. First the waste is segregated, recyclable material are sent back and rest is burnt.

• Rotatory Kilo Incinerator:

It consists of two burners

Primary ~~commercial~~ combustion chamber (rotating)
Secondary combustion chamber.

Procedure :- Both liquid and solid waste are introduced in first chamber which are burnt at $1800^\circ F$. The kiln constantly rotating to ensure best combustion efficiency. The ashes left in first burner are called fly and bottom ashes and the gaseous component ~~convined~~ are condensed and atomised through nozzle into the after-burner. The waste is burnt at $2200^\circ F$ and the bonds b/w gases also breaks down. These atoms combine with O_2 to form stable products and this unit is always attached with scrubbers and filters.

Advantage:

- The volume of waste is reduced by 75-80%.
- Fly ashes can be used for making tiles, bricks.
- The high heat of the incinerators are used to produce

electricity.

Working

Advantages

} from book.

• Composting and its Types } from book.

• Recycling and Reuse } from book.

→ NUCLEAR ACCIDENTS & HOLO CAST

Nuclear Accidents

- Lost, stolen or abandoned radio-active source.
- Human error during plant servicing, medical treatments.
- Equipment failures
- Transport accidents
- Natural calamities

Nuclear Hazard.

Caused by high amount of ionising and non-ionising radiations which leads to degradation of environment and the life on planet.

These hazards form huge clouds of radio-active particles which get transferred to distant places with winds and get deposited on the soil and water causing air pollution and through which it enters into the animal system.

effects: (Unit:- RAD → Radiation Absorbic Dose.)

- ① Somatic Effects :- More than 10 RAD for the short durations shows somatic effects like cancer, cataract, hair loss
- ② Genetic : More than 600 RAD for the short and long durations causes damages which are heritable.

Types of Dose:

- ① Acute Dose :- It has the dose at or more 10 RAD.

- 50 RAD → benine cancer.
- 100 RAD → damages bone marrow, spleen tissues
- 200 - 300 RAD → Hair loss and ~~hair loss~~ skin rashes
- 600 RAD → Sterilisation
- 1000 RAD → Damages slow dividing cells.
- 5000 RAD → Damages nerve cells and reproductive cells.

②. Chronic Dose: It is the small dose over a long period from which the cells can easily be covered. Less than 10 RAD.

Control:

- Nuclear power plants should be located away from populations. with suitable radiation absorbent zone.
- Strict safety measures
- Waste disposal should be well planned and according to the policies.
- High level waste are converted to inert solids, then disposed underground.
- Medium level waste are solidified, mixed with concrete and then disposed.
- Low level waste can be put directly into containers and disposed.

Examples (from book).

AIR POLLUTION

Pollutant

1. Carbon Monoxide :- Most abundant sources.

- Incomplete combustion of fuels.
- Industrial & automobile exhaust.
- volcanic eruptions.
- Micro-organisms in soil.
- Natural & marsh gas emission.

Effects:

- It has high affinity for haemoglobin (HB). It forms complex with HB which causes ~~Asphyxiation~~ hypoxia, headaches and laziness.

Control:

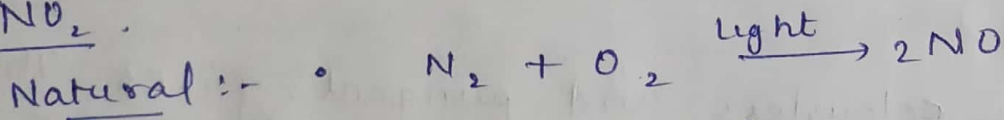
- Modification of engine design: → A low fuel to air ratio reduces the amount of NO_x but increases CO production, so high oxygen proportions supports the complete combustion and prevent its formation.
- Fuel Modification: → light weight hydrocarbons like CH_4 , CNG should be used.
- Treatment of exhaust gases: before emitting as exhaust through catalytic converters.

POLLUTANTS

I. NO_x . (NO , NO_2)

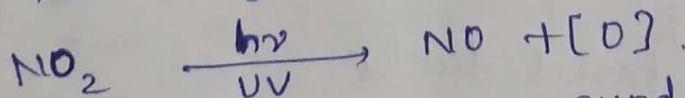
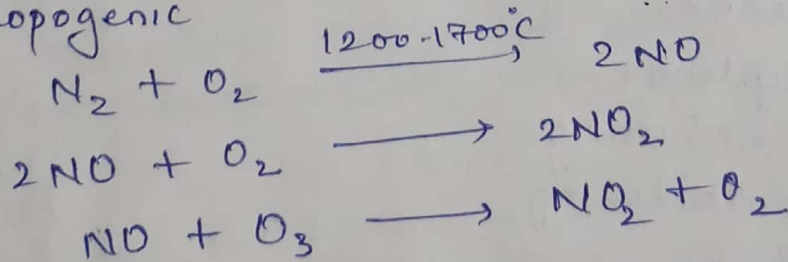
NO is a non-harmful gas but is converted into NO_2 which is harmful.

NO_2



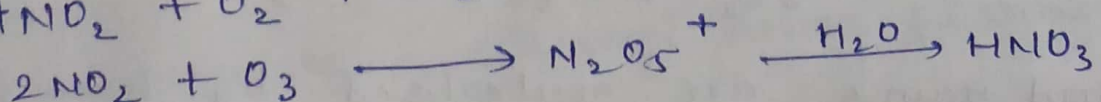
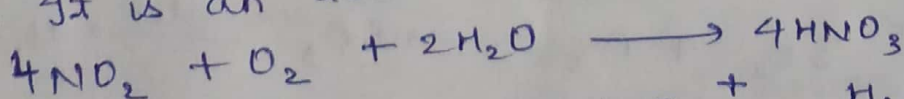
- Nitrogen Fixation: Aerobic decomposition of nitrogenous organic matter.

- Anthropogenic



$[\text{O}] \rightarrow$ Nascent oxygen. forms ^{ground} ozone layer.

Sink: It is an acid rain.



Acid rain is getting neutralised by bases like lime.

Effect:

- 1) NO_2 reacts with hydrocarbon in presence of hv to form photo-chemical smog.
- 2) formation of acid rain.
- 3) At high levels NO_2 is lung irritant and also causes bronchitis.

Control:

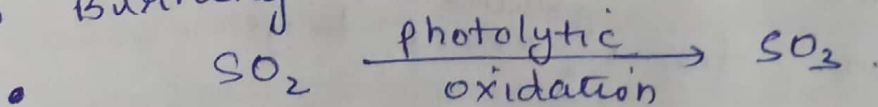
- Modification of engine design.
- If combustion takes place at low temp and high O_2 level, there is immense reduction of NO_x emission.
- Scrubbing of exhaust gases with H_2SO_4 in the scrubber.
- Using catalytic converters to convert NO_x into Nitrogen gas and Ammonia.

II. SO_x (SO_2, SO_3)

These are ~~color~~ colourless and pungent gases. They are required by plants in very small amount. If level increases they become strong air pollutant.

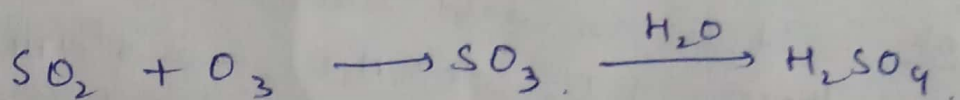
Causes: (Source).

- Volcanic eruptions.
- Petroleum refining
- Cement manufacturing
- Copper smelting.
- Burning of fossil fuels.



Sink:

- Ozone



- Acid rains are neutralised by bases.

Effects:

- cause acid rain.
- respiratory irritants
- causing Chronic bronchitis
- Claurauses in plants.
- Damage to the marble buildings.
- Rotting of protein fibres. $\rightarrow$ cotton, wool
- Papers become brittle and pale.

Control.

- Using sulphur-less fuel.
- Using chemical scrubbers. to remove SO_2
- Using citric acid which forms a citrate complex from which pure sulphur can be precipitated after passing H_2S gas.

III. Particulate Matter

Any dispersed solid or liquid matter of size $0.002\mu - 500\mu$ (dia)

- Aerosol: These are tiny solid / liquid particles.
- Dust: Fine solid particles.
- Fumes: These are the solid particles formed after condensation of vapours.
- Mist / fog.
- Smoke / Soot: Soot mainly consist of carbon induced due to incomplete combustion.
- Fly ashes:

Sources.

- Volcanic eruptions.
- Dust storms.
- Industrial processes.
- Wind
- Burning of fuels.
- Thermal power plants.

Effects

- Causing Bronchitis, lung cancer.
- Damaging the buildings and soil.
- Causing corrosion.

Control

- Using devices :- ~~Brevity~~ Gravity settling chamber, cyclonic collector, fabric filters, scrubbers, electrostatic precipitators.

IV Hydrocarbons

Low molecular weight & volatile.

Sources

- Plants.
- Anaerobic decomposition of organic matter.
- Fossil fuel exhaust.
- Evaporation from organic solvents.

Effects

- Carcinogenic.
- Produce photo-chemical smog.
- Respiratory problems.
- Methane has Narcotic effects.
- VOC's are highly toxic.

Control

- Getting oxidised and washed away.
- Adsorption by using bed of activated ~~Carbon~~ Carbon filters.
- Modification in storage tank design to prevent evaporations.

Impacts of Air Pollution

① Acid Rain.

Photochemical Smog

NO_x Generates atomic O_2



$[\text{O}]$ generates free radicals.



Free radical produces HC radicals



HC Radiation Generates

HC peroxide / Peroxy Radical



Aldehydes / aldehyde peroxide



It forms peroxy acetyl Nitrate (PAN)

(PAN + O_3 + HC + Aldehydes + Ketones)

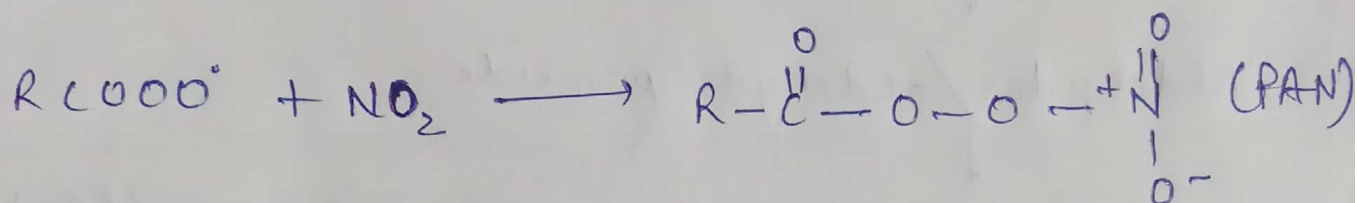
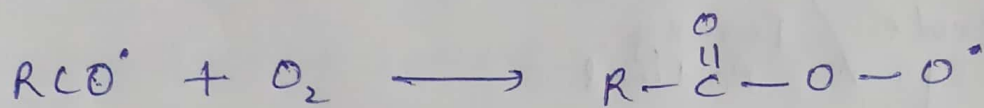
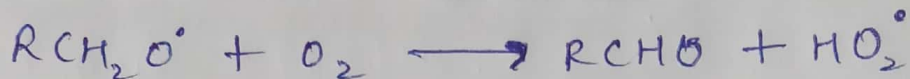
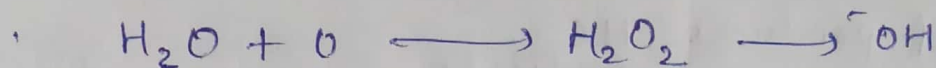
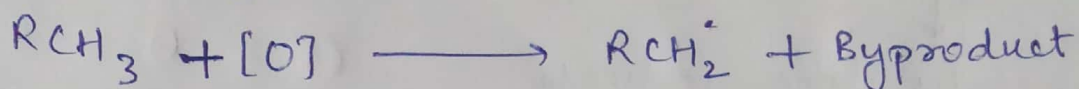
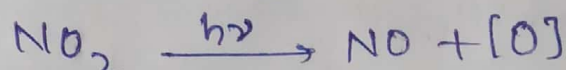
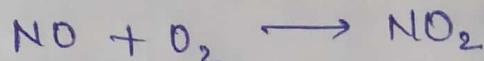
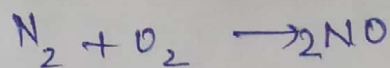
I. London Smog

SO_x + Humidity + Fog + Particulates.

It is London ~~smog~~ ^{smog} which killed 4000 people in London, 1952. Along with respiratory problems and bronchitis in the population. After sunrise its amount increases due to fast conversion of SO_2 into SO_3 , and the droplets of H_2SO_4 get deposited on the carbon particles and forms the smog.

II. Los-Angeles Smog

It is a haze formed due to photochemical reaction when the air has NO_x , hydrocarbons in presence of bright sunlight and temp $> 18^\circ\text{C}$. It has oxidising nature. The worst ingredient is ground level ozone and pan.

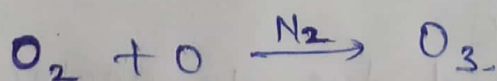
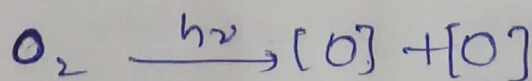


GREENHOUSE EFFECT

- Doubling of greenhouse gases leads to the increase in temp. of earth by 5 to 6°C.
- It lowers down the food production.
- Regional climate change.

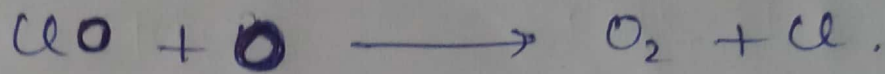
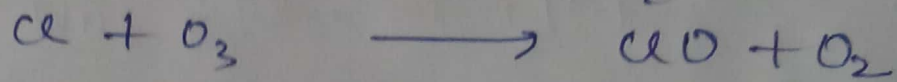
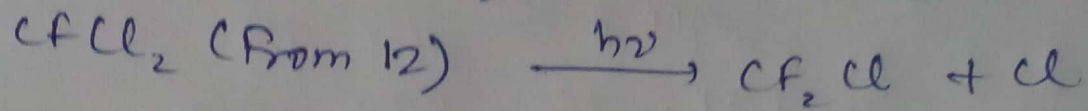
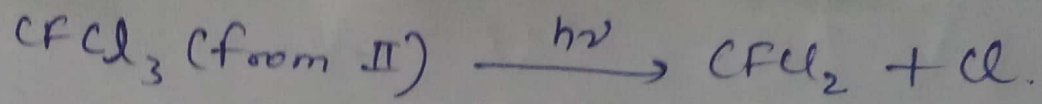
Ozone Layer Depletion

90% of ozone forms a thick blanket in the stratosphere in ozone layer which blocks U.V. radiations.



CFCs, sources:

- Coolant of refrigerator.
- Propellant in aerosol sprays.



Equipments :

- Gravitation settling chamber.
- Cyclo-tronic chamber.